

PALLETPILOT · DESIGN DECISIONS REGISTER

PalletPilot design decisions register

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DOCUMENT CONTROL

Document ID	PLP-DEC-001	Version	0.2
Status	Draft	Date	2026-10-01
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/palletpilot		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Amish accepted the recommendation of open item 1 (steering range, option a, with a TRL 4 aisle test as a go or no-go gate; PLP-DDR-003 A1); moved to decisions made; open items renumbered 1 to 5
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions from PLP-DDR-001 to 003 and the review note; budget treated as a value-engineering target

PalletPilot design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Kit mass of 44.4 kg against R15's 40 kg (2.0 kg over was accepted for now)	(a) Accept for the prototype and recheck at the motor quote; (b) lighten now (5 mm top plate, tube lever, hollow pivot pin, about 1.5 kg)	(a), keeping (b) for the TRL 4 drawings	Plate thickness, lever and pin stock	PLP-DDR-003, A2
2	Drive wheel lift of 10 mm with the release lever raised	(a) Accept for smooth indoor floors; (b) lower the cam shaft and raise the hangers for about 15 mm	(a)	Cam throw, hanger height	PLP-DDR-003, A3
3	Classification of follow mode under ISO 3691-4 and the safety functions it then needs	Classify as a driverless truck function, or as an operator-controlled truck with an assist mode	None yet	Not part of the TRL 3 build; sets the stop functions and sensor rating needed later	PLP-DDR-001, O2
4	Donor jack models and steering yokes to support first	Survey three common 27 x 48 in, 2,500 kg jacks from published drawings	None yet	Packing bar thickness, clamp and notch, steering stop angle	PLP-DDR-001, O3
5	Named site and co-design partners (types decided: one small warehouse, one maker space)	Picked per area later	None yet	None in the build	PLP-DDR-001, O1

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The donor jack's yoke plate thickness, rear edge position and width, the pump diameter, and the distance from the steering axis to the frame head	They set the packing bar, the clamp holes, the notch and the steering stop angle	PLP-DDR-003; R1
2	The hub motor's 24 V winding, peak torque (30 N·m or more), brake holding torque (20 N·m or more), mass and its shaft's diameter and flats	They set the drive arm's axle hole and the mass and stopping figures	PLP-DDR-002; PLP-CAL-001
3	The preload spring: about 34 mm outside diameter, about 65 mm free length, about 100 N/mm	It sets the 750 N per wheel preload and the 10 mm wheel lift	PLP-DDR-003; PLP-CAL-001 [G4]
4	The safety edge profile's travel (40 mm or more at the front), its evaluation unit's category and response time	R8's 0.15 m/s limit rests on 40 mm of travel and a 100 ms chain	PLP-DDR-002; R8
5	The tiller head fits the donor handle tube and carries the beacon, stop and anchor	The head clamps on the handle; the grip width is still open in the appearance model	REVIEW (2026-09-26)
6	The lidar's mounting holes and scan plane height above its base	The bracket shelf is drilled to suit and sets the 200 mm scan plane	PLP-DDR-003

■ Value engineering

Value-engineering target: USD 1,610 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,690 (USD 80 over the target). Main cost drivers and savings worth trying:

- The largest lines are the two hub motors (USD 400), the battery (USD 180), the motor driver (USD 150), the drive module (USD 130) and the contact bumper (USD 110); together they are about 57 % of the kit.
- Making the design constructable added USD 80: the drive module with its clamp, arms and pivot (USD 90 to USD 130), the release mechanism with its cams, link, springs and straps (USD 20 to USD 45), the enclosure feet (USD 5) and the steering stops (USD 10).
- Savings worth trying: a motor sold with its driver as a pair (the class prices run from USD 165 to USD 249 per motor), batching the laser-cut plate parts into one order, and a tube lever and hollow pivot pin, which also save mass.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D9: layered stopping with a low-cost lidar layer, budget, dual hub motors on a sprung yoke module, 24 V LiFePO4, everything on the yoke, speeds, kit rating, legal route, partner types	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	PLP-DDR-001
2026-09-25	Bumper-only speed limited to 0.15 m/s (O4); mass and cost overruns accepted for now, R5 fixes evaluated at TRL 4 (O5); a second main contactor in series (O6)	Amish: "i accept all your recommendations, go with them across all repos."	PLP-DDR-002
2026-09-26	Budget set to USD 1,610 to cover the priced BOM	Amish: "i approve all the budget items."	PLP-DDR-002 v0.2
2026-09-30	Make the design physically buildable while drawing the build plan; outstanding decisions go in this register, not in the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	PLP-DDR-003 (Draft, open for review)
2026-10-01	The budget is a value-engineering target, reported as over or under, never as a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	This register; PLP-CAL-001 v0.4
2026-10-01	Steering range (open item 1 in register v0.1): option (a). Accept about 40° each way with the kit fitted for the prototype (a bare jack turns about 90°), turning radius roughly 1.5 m about the load wheels; set the rubber stops from the donor survey; the precis, README and requirements no longer say the kit steers like the bare jack. Go or no-go gate: the TRL 4 aisle test must show the kit can make a right-angle turn into a standard pallet bay; if it cannot, the drive layout changes before TRL 5 (option c, for example one centre drive wheel under the yoke, which reopens D3)	Amish: "i accept your recommendations for PalletPilot"	PLP-DDR-003, A1